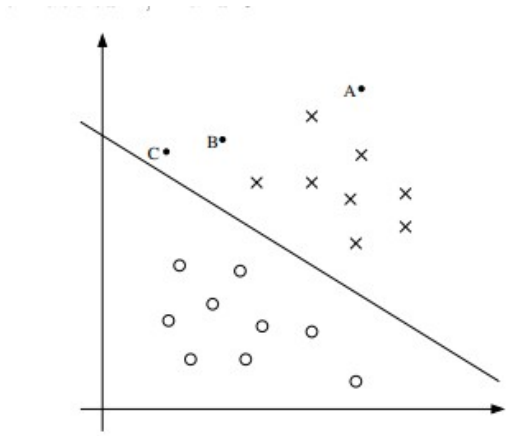
Homework One

Support Vector Machines and Phase Transition of Ising Model

1. Theory: Support Vector Machines

1.1. Margins and our optimal problem

Let us first gain some visual intuition in a problem where there are only two features for our data like the picture below taken from Andrew Ng's notes.



For a simple numerical example, we can consider the separating hyperplane to be

$$x_1 + 2x_2 - 2 = 0 \tag{1}$$

Then we can verify the crosses that we will label as $y = +1$ has $x_1 + 2x_2 - 2 > 0$ (which is a very specific choice of $w^T x + b > 0$ for a specific training data point) and the circles that we will label as $y = -1$ has $x_1 + 2x_2 - 2 < 0$. The closer the points are to the hyperplane, the smaller the magnitude $x_1 + 2x_2 - 2$ is.

With this intuition, we will formalize the notation of the functional margin $\hat{\gamma}^{(i)}$ of (w, b) (representing a hyperplane) with respect to the training example $(x^{(i)}, y^{(i)})$. Note that an upper index in parentheses denotes a specific training sample.

$$\hat{\gamma}^{(i)} = y^{(i)}(w^T x^{(i)} + b) \tag{2}$$

a) Check that for both labels $y = 1$ or $y = -1$, our prediction is correct if $\hat{\gamma}^{(i)} > 0$.

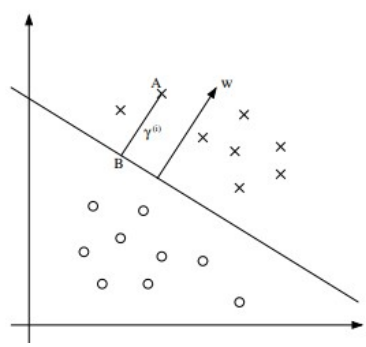
b) Check that the correctness of our prediction is not affected by a scaling of (w, b) to $(2w, 2b)$ (for example).

For a training set $S = \{(x^{(i)}, y^{(i)}); i = 1, \ldots, n\}$, we define the functional margin respect to the set as the smallest of the function margin of the individual samples

$$\hat{\gamma} = \min_{i=1,\ldots,n} \hat{\gamma}^{(i)} \tag{3}$$

As we can see from the previous picture, we are more likely to label A correctly and may have trouble labeling C correctly. So the nearest data point to the hyperplane is more important.

Now let us define the geometric margins.



In the above picture we also show vector w along with the hyperplane.

c) Show that the vector w is orthogonal to the hyperplane (it is sufficient to show it in the 2D case).

In the picture data point A (with location $x^{(i)}$) is a distance of AB away from the hyperplane. This distance is the geometric margin, which we denote $\gamma^{(i)}$ (with no hat).

d) What is the location of B in terms of w, $||w||$, $x^{(i)}$ and $\gamma^{(i)}$?

e) Use the fact point B is on the hyperplane to derive

$$\gamma^{(i)} = y^{(i)} \left(\left(\frac{w}{||w||} \right)^T x^{(i)} + \frac{b}{||w||} \right) \tag{4}$$

Note that when $||w|| = 1$, the functional and geometric margins agree. We also define the geometric margin respect to the set as

$$\gamma = \min_{i=1,\ldots,n} \gamma^{(i)} \tag{5}$$

The intuition is that if a decision boundary (hyperplane) maximize the margin, it reflects a very confident set of predictions. Assuming we are working with a training set that is linearly separable, the optimization problem we are going to work on is

$$\begin{aligned} &\max_{\gamma,w,b} \gamma \\ &\quad \text{s.t.} = y^{(i)}(w^T x^{(i)} + b) \geq \gamma, i = 1, \ldots, n \\ &\quad ||w|| = 1 \end{aligned} \tag{6}$$

or phrased with functional margin as $\gamma = \frac{\hat{\gamma}}{||w||}$,

$$\begin{aligned} &\max_{\hat{\gamma},w,b} \frac{\hat{\gamma}}{||w||} \\ &\quad \text{s.t.} = y^{(i)}(w^T x^{(i)} + b) \geq \hat{\gamma}, i = 1, \ldots, n \end{aligned} \tag{7}$$

Further more, as we show scaling (w, b) will give us the same geometric margin, we can use this freedom to scale $\hat{\gamma} = 1$. Notice maximizing $\frac{\hat{\gamma}}{||w||}$ is the same thing as minimizing $||w||$ (since $\hat{\gamma} = 1$) which is the same thing as minimizing $\frac{1}{2}||w||^2$ so our optimizing problem becomes

$$\begin{aligned} &\min_{w,b} \frac{1}{2}||w||^2 \\ &\quad \text{s.t.} = y^{(i)}(w^T x^{(i)} + b) \geq 1, i = 1, \ldots, n \end{aligned} \tag{8}$$

This can be solved by commercial quadratic programming code. But let us try to solve the problem using Lagrangian multipliers.

1.2. Lagrangian duality

Consider a problem of the following form

$$\begin{aligned} &\min_w f(w) \\ &\text{s.t.} h_i(w) = 0, i = 1, \ldots, l \end{aligned} \tag{9}$$

We can define the Lagrangian to be

$$\mathcal{L}(w, \beta) = f(w) + \sum_{i=1}^{l} \beta_i h_i(w) \tag{10}$$

where β_i's are called Lagrangian multipliers and the optimization problem is solved by taking the partial derivatives to be 0

$$\frac{\partial \mathcal{L}}{\partial w_i} = 0; \frac{\partial \mathcal{L}}{\partial \beta_i} = 0 \tag{11}$$

and solve for w and β.

Now we are going to generalize this method to inequality constraints as our margin optimization problem has. Consider the following primal optimization problem

$$\begin{aligned} &\min_w f(w) \\ &\text{s.t.} g_i(w) \leq 0, i = 1, \ldots, k \end{aligned} \tag{12}$$

To solve it let us consider the Lagrangian

$$\mathcal{L}(w, \alpha) = f(w) + \sum_{i=1}^{k} \alpha_i g_i(w) \tag{13}$$

where α_i's are the Lagrangian multipliers.

Consider the quantity

$$\theta_P(w) = \max_{\alpha_i \geq 0} \mathcal{L}(w, \alpha) \tag{14}$$

where P stands for primary.

a) Verify that $\theta_P(w)$ is $f(w)$ if w satisfies primal constraints and ∞ otherwise.

Then our original problem is the same as $\min_w \theta_P(w) = \min_w \max_{\alpha_i \geq 0} \mathcal{L}(w, \alpha)$. We denote the optimal value as p^*.

Now let us consider a slightly different problem, denote

$$\theta_D(\alpha) = \min_w \mathcal{L}(w, \alpha) \tag{15}$$

where D stands for dual. In the primary problem we are maximizing respect to α_i and now we are minimizing respect to w. Then our dual optimization problem is to find

$$\max_{\alpha \geq 0} \theta_D(\alpha) = \max_{\alpha \geq 0} \min_w \mathcal{L}(w, \alpha) \tag{16}$$

We denote the optimal value for this problem $d^* = \max_{\alpha \geq 0} \theta_D(\alpha)$.

b) Show that $d^* \leq p^*$.

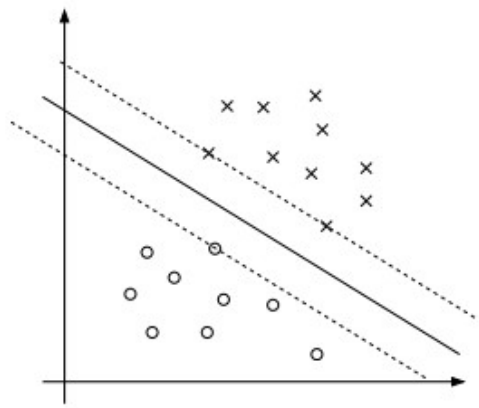
Under the condition that f and g_i's are convex and constraints g_i are strictly feasible (there exist some w, so that $g_i(w) < 0$ for all i). We have $d^* = p^* = \mathcal{L}(w^*, \alpha^*)$ where w^* is the solution for the primary problem and α^* is the solution of the dual problem. The solution satisfies the so called Karush-Kuhn-Tucker (KKT) conditions

$$\begin{aligned} \frac{\partial \mathcal{L}(w^*, \alpha^*)}{\partial w_i} &= 0, i = 1, \ldots, d \\ \alpha_i^* g_i(w^*) &= 0, i = 1, \ldots, k \\ g_i(w^*) &\leq 0, i = 1, \ldots, k \\ \alpha^* &\geq 0, i = 1, \ldots, k \end{aligned} \tag{17}$$

The first line is the usual condition, the third line is the constraint. The condition on α imposed so that the primary optimization problem has a solution (otherwise we can choose α to be negative infinity to maximize $\mathcal{L}$, clearly not what we want.)

c) What does $\alpha_i^* g_i(w^*) = 0$ mean if $\alpha_i > 0$? What does that mean about our data set?

We call the data set that saturates the constraint "support vectors". SVM only have a small number of "support vectors".



1.3. Optimal margin classifiers

Now let us get back to our margin optimization problem,

$$\begin{aligned} &\min_{w,b} \frac{1}{2}||w||^2 \\ &\text{s.t.} = y^{(i)}(w^T x^{(i)} + b) \geq 1, i = 1, \ldots, n \end{aligned} \tag{18}$$

a) Write the constraints in the form of $g_i(w) \leq 0$.

b) What is the Lagrangian $\mathcal{L}(w, b, \alpha)$?

   Let us find the dual form of our optimization problem, for which we need to minimize the Lagrangian with respect to w and b for fixed α.

c) Solve $\nabla_w \mathcal{L}(w, b, \alpha) = 0$ and $\frac{\partial \mathcal{L}(w,b,\alpha)}{\partial b} = 0$. At this step you should obtain

$$w = \sum_{i=1}^{n} \alpha_i y^{(i)} x^{(i)} \tag{19}$$

$$\sum_{i=1}^{n} \alpha_i y^{(i)} = 0$$

d) Now plug these results back into the Lagrangian and obtain

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)}. \tag{20}$$

Then the dual optimization problem we have is

$$\begin{aligned} &\max_\alpha \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} \\ &s.t.\ \alpha_i \geq 0, i = 1, \ldots, n \\ &\sum_{i=1}^{n} \alpha_i y^{(i)} = 0 \end{aligned} \tag{21}$$

Suppose we are able to solve this dual problem, then we will also obtain the solution to the original problem w^*.

e) Now consider the primary problem, and show that the optimal value for the interception is

$$b^* = -\frac{\max_{i:y^{(i)}=-1} w^{*T} x^{(i)} + \min_{i:y^{(i)}=1} w^{*T} x^{(i)}}{2} \tag{22}$$

Now if we want to make a prediction, we would use our solution to calculate $w^T x + b$.

f) Show that we can put it in a form that it only depends on the inner products between x and the support vectors.

2. Numerical: Classify the Phases

In this assignment, we continue the problem of classifying the phases of the Ising Model. You need to submit the written answers to the questions, the plot you make, and the codes you wrote to generate the plots and other codes that you implemented to perform your analysis. The codes need to be in a file that can directly run on symmetry and other files need to be in pdf format.

a) In the tutorial, we used accuracy as the scoring method. In this assignment, you should also use precision, recall, and F1-score to evaluate.

b) We know the temperature of the data. Make a plot of the average accuracy as a function of the temperature. Where does our model perform well? Where does our model perform badly? Could you use your knowledge of the Ising model to explain this? What kind of feature do you think we can add to improve the performance (you don't need to implement this).

c) We have also learned how to use the support vector machine to classify. Use tutorial two's codes as template, train a support vector machine code (what kernel should we use? If unsure, use both. Complicated kernels will take a few minutes to run.) to classify the data. Also make a plot of the average accuracy as a function of the temperature. Compare the two methods: Logistic regression v.s. support vector machine. Which is better? Why?

A. References

The theory part is written base on Andrew's notes.